



Bode plot

Bode plot is a plot of the magnitude of the transfer function in decibels and the phase of the transfer function in degrees, versus frequency in logarithmic scale on a semilog graph.

Bode plots can often be approximated by straight lines and are easy to draw.

Bode plot of the first order low pass filter

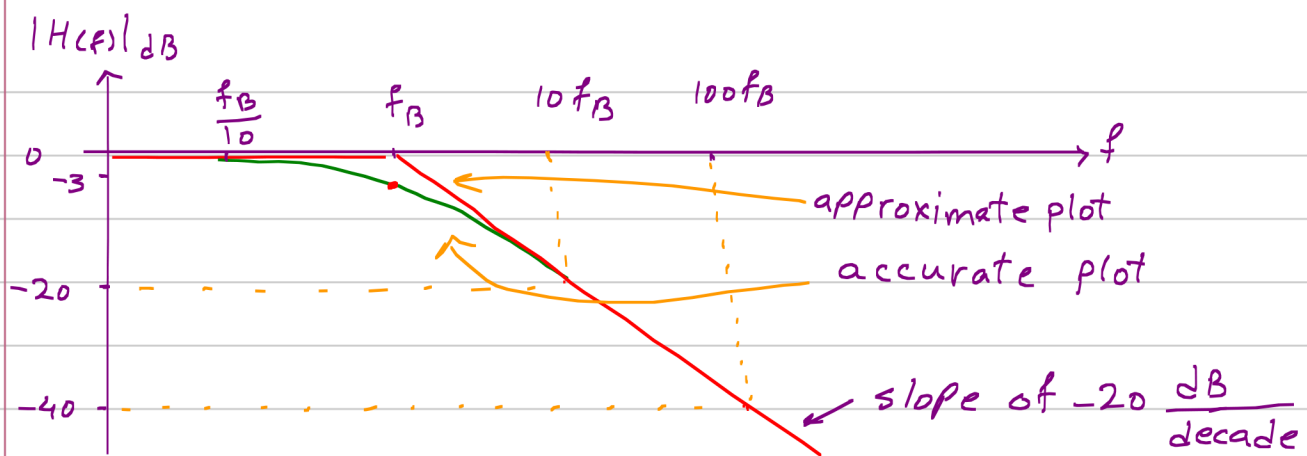
$$H(f) = \frac{1}{1 + j \frac{f}{f_B}} \Rightarrow |H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_B}\right)^2}}$$

$$|H(f)|_{dB} = 20 \log \frac{1}{\sqrt{1 + \left(\frac{f}{f_B}\right)^2}} = \cancel{20 \log 1} - 20 \log \sqrt{1 + \left(\frac{f}{f_B}\right)^2}$$

$$\text{at } f \ll f_B \Rightarrow |H(f)|_{dB} \approx 0$$

$$\text{at } f = f_B \Rightarrow |H(f)|_{dB} = -3 \text{ dB}$$

$$\text{at } f \gg f_B \Rightarrow |H(f)|_{dB} \approx -20 \log \frac{f}{f_B} \quad \left(\text{straight line with a slope of } -20 \frac{\text{dB}}{\text{decade}} \right)$$

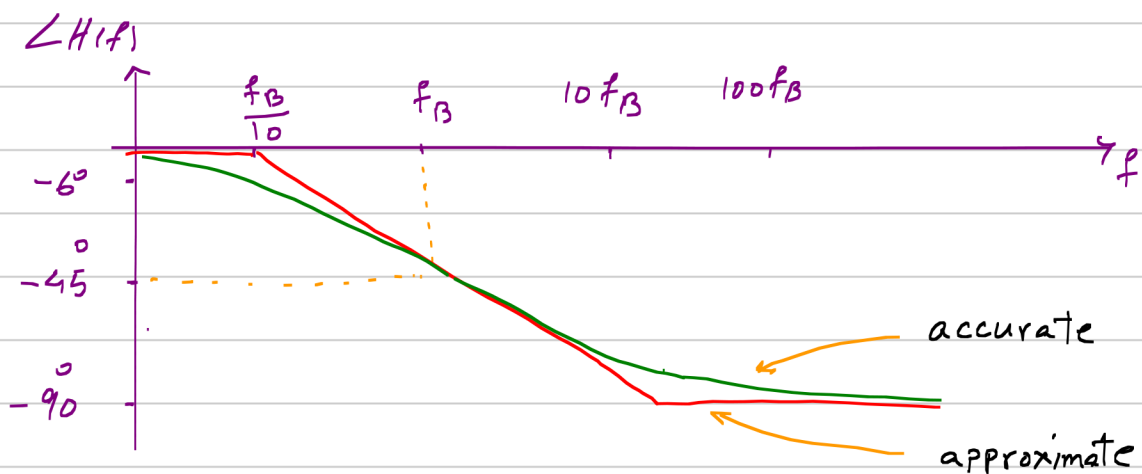


Phase plot: $\angle H(f) = -\tan^{-1} \frac{f}{f_B}$

at $f \ll f_B$: $\angle H(f) \approx 0$ (at $f = \frac{f_B}{10}$, $\angle H = -6^\circ$)

at $f = f_B$: $\angle H(f) = -45^\circ$

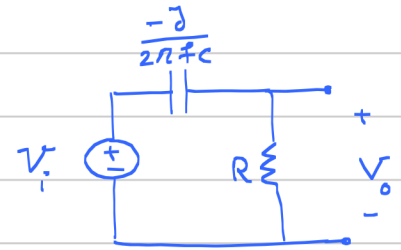
at $f \gg f_B$: $\angle H(f) \approx -90^\circ$ (at $f = 10f_B$, $\angle H = -84^\circ$)



First order RC high pass filters

$$H(f) = \frac{V_o}{V_i} = \frac{R}{R - \frac{j}{2\pi fC}}$$

$$= \frac{1}{1 - \frac{j}{2\pi fRC}} = \frac{2\pi fRC}{2\pi fRC - j1} \times \frac{j}{j} = \frac{j2\pi fRC}{j2\pi fRC + 1}$$



$$H(f) = \frac{j \frac{f}{f_B}}{1 + j \frac{f}{f_B}}$$

$$\text{where, } f_B = \frac{1}{2\pi RC}$$

$$\Rightarrow |H(f)| = \frac{\frac{f}{f_B}}{\sqrt{1 + \left(\frac{f}{f_B}\right)^2}}, \quad \angle H(f) = 90 - \tan^{-1} \frac{f}{f_B}$$

straight line with slope of $20 \frac{\text{dB}}{\text{dec}}$

$$\text{If } f \ll f_B \Rightarrow |H(f)|_{\text{dB}} = 20 \log \frac{f}{f_B}, \quad \angle H(f) = 90^\circ$$

$$\text{If } f = f_B \Rightarrow |H(f)|_{\text{dB}} = 20 \log \frac{1}{\sqrt{2}} = -3 \text{ dB}, \quad \angle H(f) = 45^\circ$$

$$\text{If } f \gg f_B \Rightarrow |H(f)|_{\text{dB}} = 20 \log 1 = 0, \quad \angle H(f) = 0$$

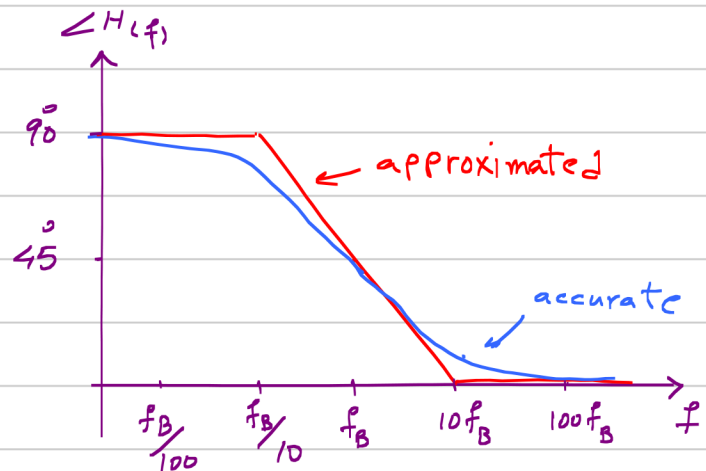
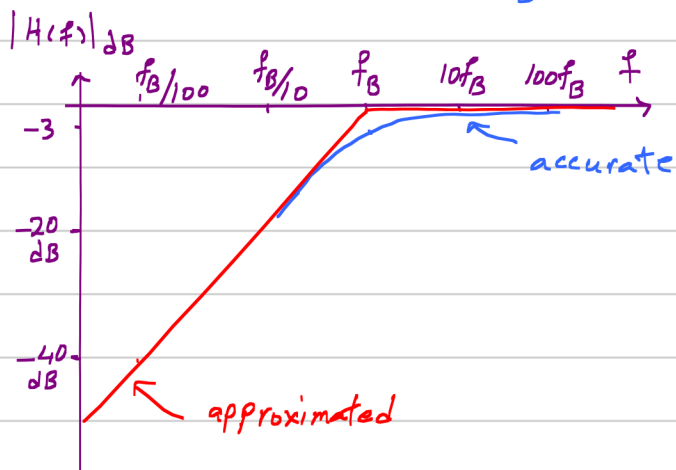


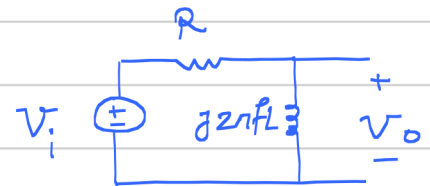
Table 6.2. Transfer-Function Magnitudes and Their Decibel Equivalents

$ H(f) $	$ H(f) _{dB}$
100	40
10	20
2	6
$\sqrt{2}$	3
1	0
$1/\sqrt{2}$	-3
1/2	-6
0.1	-20
0.01	-40

$$|H(f)|_{dB} = 20 \log |H(f)|$$

Ex. Find the type of filter. what is the break frequency?

$$H = \frac{V_o}{V_i} = \frac{j2\pi fL}{R + j2\pi fL} = \frac{j2\pi f \frac{L}{R}}{1 + j2\pi f \frac{L}{R}}$$



$$H(f) = \frac{j \frac{f}{f_B}}{1 + j \frac{f}{f_B}}, \quad f_B = \frac{R}{2\pi L}$$

This is the transfer function of a high pass filter with the break frequency of $\frac{R}{2\pi L}$.

It is also possible to find the type of filter using this reasoning:

$$Z_L = j2\pi fL \Rightarrow \text{at low frequency } (f \approx 0) \quad Z_L \approx 0 \Rightarrow$$

inductor is close to a short ckt. $\Rightarrow V_o = 0$

$\Rightarrow$ NO low frequency components exist at the output.

at high freq., Z_L is large (close to open ckt.) $\Rightarrow V_o = V_i$

$\Rightarrow$ High frequency component can pass through the filter

Therefore, this is a high pass filter.

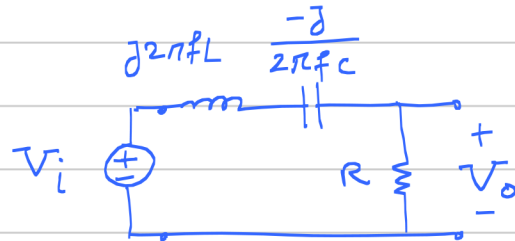
Band pass filter

In some applications it is desired to select a certain range of frequencies and discard the other frequencies. Such filters are useful in radio receivers for example.

This filter is called band pass filter and can be made using RLC circuit. (second order circuits)

Total impedance is

$$Z_s = R + j2\pi fL + \frac{-j}{2\pi fC}$$



Resonant frequency is defined as the frequency at which the

total impedance is purely resistive. ($Z_L + Z_C = 0$). Therefore at

resonant frequency f_0 :

$$j2\pi f_0 L - \frac{j}{2\pi f_0 C} = 0 \Rightarrow 4\pi^2 f_0^2 LC = 1 \Rightarrow$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{resonant frequency}$$

Quality factor is defined as :

$$Q_s = \frac{|Z_L|}{R} \Big|_{f=f_0} = \frac{2\pi f_0 L}{R} \quad (1), \quad \text{Since } |Z_L| = |Z_C| \text{ at } f=f_0, Q_s \text{ is also;}$$

$$Q_s = \frac{|Z_C|}{R} \Big|_{f=f_0} = \frac{1}{2\pi f_0 RC} \quad (2), \quad \text{Then the total impedance can be written as:}$$

$$Z_s = R \left[1 + jQ_s \left(\frac{f}{f_0} - \frac{f_0}{f} \right) \right]$$

multiplied by Q_s from (1) multiplied by Q_s from (2)

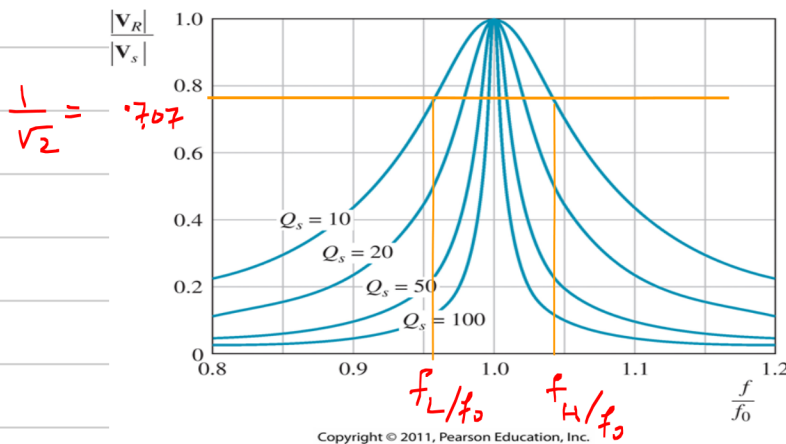
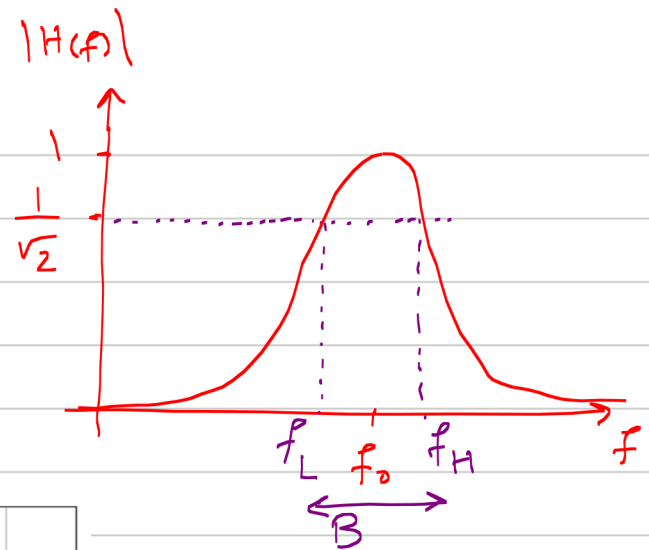
$$H(f) = \frac{V_o}{V_s} = \frac{R}{Z_s} = \frac{1}{1 + jQ_s \left(\frac{f}{f_0} - \frac{f_0}{f} \right)}$$

$$|H(f)| = \frac{1}{\sqrt{1 + Q_s^2 \left(\frac{f}{f_0} - \frac{f_0}{f} \right)^2}}$$

at $f = f_0$, $|H(f)| = 1$

at $f \ll f_0$, $|H(f)| \approx 0$

at $f \gg f_0$, $|H(f)| \approx 0$



Higher quality factor results

in narrower band width.

$$B|_{Q_s=10} = f_H - f_L$$

Half power frequencies are defined as the frequencies at which the

transfer function magnitude has fallen from its maximum by a factor

of $\frac{1}{\sqrt{2}} = 0.707$. For any quality factor Q_s , there are two half-power

frequencies f_L and f_H . The difference between these two

frequencies is called bandwidth . $B = f_H - f_L$

In a bandpass filter, bandwidth contains the range of frequencies that

will pass through the filter. It can be shown that $B = \frac{f_0}{Q_s}$

Band stop filter (notch filter)

In some applications it is desired to discard a certain range of frequencies and pass the other frequencies. This filter is called band stop filter and can be made using RLC circuit. (second order circuits)

$$H(f) = \frac{V_o}{V_i} = \frac{j2\pi fL + \frac{-j}{2\pi fc}}{R + j2\pi fL + \frac{-j}{2\pi fc}}$$

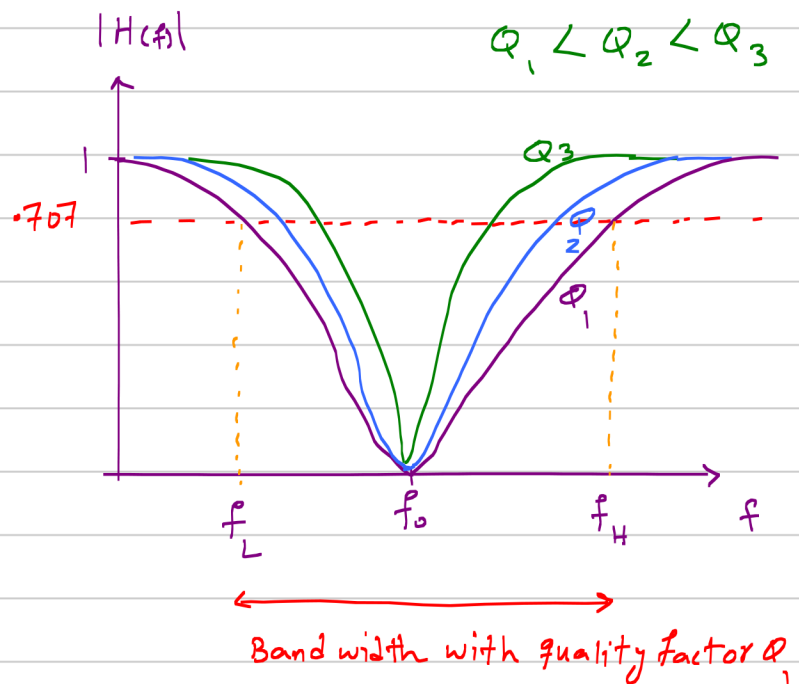
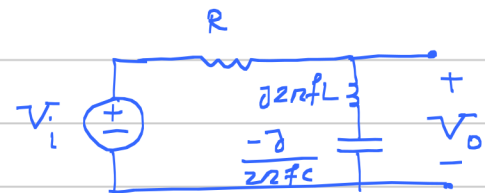
$$= \frac{jRQ_s \left(\frac{f}{f_0} - \frac{f_0}{f} \right)}{R \left(1 + jQ_s \left(\frac{f}{f_0} - \frac{f_0}{f} \right) \right)}$$

$$|H(f)| = \frac{Q_s \left(\frac{f}{f_0} - \frac{f_0}{f} \right)}{\sqrt{1 + Q_s^2 \left(\frac{f}{f_0} - \frac{f_0}{f} \right)^2}}$$

at $f = f_0$, $|H(f)| = 0$

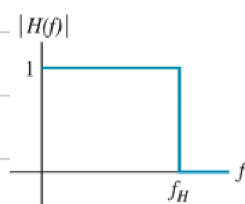
at $f \gg f_0$, $|H(f)| \approx 1$

at $f \ll f_0$, $|H(f)| \approx 1$

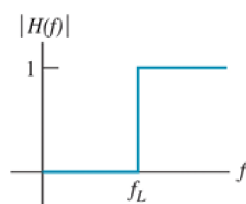


Ideal filters

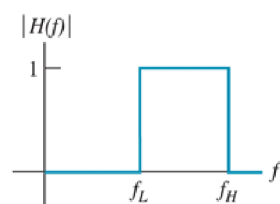
An ideal filter passes the components in the desired range with no change of amplitude or phase and totally rejects the undesired frequencies.



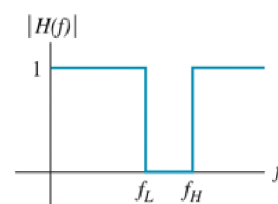
(a) Lowpass



(b) Highpass



(c) Bandpass



(d) Band reject